

MUKHAMAD BAHRUL ULUM

Automated Data Entry & Web Scraping Specialist | Python & AI Automation

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PROFESSIONAL SUMMARY

Results-driven Automated Data Entry & Web Scraping Specialist with a strong analytical foundation in Accounting and Business Administration. I leverage Python automation, Selenium, and AI orchestration to extract, clean, and input massive datasets at 10× the speed of manual work with zero human error. I transform messy, unstructured web data and documents into clean, structured, business-ready spreadsheets and databases — saving clients hundreds of manual hours.

CORE COMPETENCIES

✓ Large-Scale Web Scraping & Data Mining ✓ Advanced Automated Data Entry ✓ AI-Powered Data Cleaning & Formatting
✓ Lead Generation Pipelines ✓ OCR Document Processing ✓ Python Automation Scripting ✓ LLM / AI Integration

TECHNICAL SKILLS

Web Scraping & Crawling	Selenium, BeautifulSoup, Scrapy, Requests
Data Entry & Automation	Python Automation, Pandas, OpenPyXL, pyautogui
AI & LLM Integration	Prompt Engineering, LLM Pipelines, OCR (Tesseract)
Backend & APIs	FastAPI, Python 3.12, Node.js, REST APIs, JWT
Data Storage & Output	PostgreSQL, Redis, Excel/Google Sheets, MinIO
DevOps & Infrastructure	Docker, Nginx, Cloudflare, cPanel, HMAC-SHA256

KEY SERVICES & EXPERTISE

► **Large-Scale Web Scraping & Data Mining:** Build custom Python crawlers (Selenium + BeautifulSoup) to extract complex data from e-commerce platforms, B2B directories, real estate listings, and JavaScript-heavy dynamic websites at scale.

► **Advanced Automated Data Entry:** Eliminate manual copy-pasting by scripting automated pipelines that process, format, and upload thousands of rows into Excel, Google Sheets, or client CRMs — seamlessly and error-free.

► **AI-Powered Data Cleaning & Formatting:** Use Pandas and LLM integrations to deduplicate records, standardize formats, fix messy text, and validate datasets to 100% accuracy before delivery.

► **Lead Generation Pipelines:** Extract and enrich B2B contact lists (emails, job titles, LinkedIn profiles) from directories and social platforms for sales teams.

► **OCR Data Pipelines:** Convert scanned documents, images, and unstructured PDFs into clean, structured text databases using Tesseract OCR and custom pre-processing pipelines.

PORTFOLIO PROJECTS

Tander — Government Tender Scraper & Data Intelligence Dashboard · Python · Selenium · Pandas · LLM · React · Node.js

- Built a highly scalable multi-target web scraper (LPSE portals) for mass government procurement data extraction across Indonesia.
- Automated structured parsing of procurement documents — extracting project names, budgets, deadlines, and qualification requirements into clean dashboards.
- Integrated LLM to auto-summarize massive tender documents into concise qualification checklists for bidders.
- Engineered real-time multi-channel alert pipelines (Telegram, WhatsApp, Email) for new tenders, deadline reminders, and competitor price intelligence.

RBV CLEAN — OCR & PDF Data Extraction Pipeline · Python · CustomTkinter · Tesseract OCR · OpenCV · ocrmypdf

- Developed an advanced desktop OCR application to extract and clean data from scanned university module PDFs (including watermark detection and removal).
- Engineered a custom image pre-processing pipeline (gray-range thresholding, morphological operations) to maximize OCR accuracy on low-quality scans.
- Output: clean, structured text databases ready for downstream data entry or indexing.

Automated HR Document & Scoring Pipeline — TalentScan · Python · FastAPI · AI Integration · SMTP/IMAP

- Built automation to parse incoming applicant CVs directly from email inboxes, extracting structured candidate data into a central database.
- Automated generation of formatted scoring reports and official PKWT employment contracts, eliminating manual document preparation.

AiDiskusi — Academic PDF Data Structuring Backend · Python · FastAPI · LLM API · PostgreSQL

- Architected backend pipelines to parse, categorize, and structure thousands of academic PDF modules into organized, searchable databases.
- Automated extraction of key content segments, enabling AI-powered query matching at scale.

PROFESSIONAL EXPERIENCE

Independent Full-Stack Developer & Automation Engineer

Self-Employed / Freelance | 2026 – Present

- Design and deploy end-to-end Python automation and web scraping solutions for clients across data entry, lead generation, and document processing use cases.
- Build AI-powered data pipelines integrating LLM APIs to clean, validate, and structure large unstructured datasets.
- Deliver production-grade projects encompassing OCR pipelines, multi-target scrapers, and automated document generation systems.
- Combine accounting/business analysis background to ensure data outputs meet strict accuracy and auditability standards.

EDUCATION

Bachelor of Business Administration

Universitas Terbuka Indonesia | 2026 – Present

Academic Background in Accounting

Focus: systematic ledgers, financial auditing compliance, structural data classification — directly applicable to data quality and validation work.

LANGUAGES

Indonesian: Native / Mother Tongue

English: Professional Working Proficiency

AVAILABILITY & REMOTE WORK

Available immediately for remote freelance or contract roles. Experienced working asynchronously across time zones. Deliver clean, documented, production-ready automation scripts with full handover documentation.